

Area 4: Economic Theory and Quantitative Methods

Private Health Insurance and Healthcare Utilization: New Causal Evidence from Brazil

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Abstract

This study provides new causal evidence on the effects of private health insurance on healthcare utilization in Brazil, considering both the extensive margin (seeking medical care) and the intensive margin (number of medical consultations). Using microdata from the 2019 Brazilian National Health Survey (PNS), we exploit exogenous variation in health insurance coverage across occupational groups as an instrumental variable for the endogenous treatment of having private health insurance. Our identification strategy relies on differences in health insurance diffusion across occupation-based groups defined by economic activity, employment formalization, and state of residence, capturing institutional and labor-market-related access to private health insurance. Controlling for demographic and socioeconomic characteristics, chronic health conditions, and regional factors, the results indicate that private health insurance substantially increases healthcare utilization among employed individuals. In the intensive margin, the estimates suggest that having private health insurance more than doubles the annual number of medical consultations, with results remaining robust across alternative specifications, transformations of the dependent variable, and survey designs. The findings also reveal important heterogeneous effects. The impact of private health insurance on medical consultations is significantly larger among men, individuals without chronic diseases, and younger individuals. In particular, the marginal effect of health insurance declines with age, suggesting that private coverage generates stronger utilization gains among groups with lower baseline use of healthcare services. Regarding the extensive margin, private health insurance moderately increases the probability of seeking medical care, but substantially increases the intensity of healthcare utilization conditional on access.

Keywords: Health Insurance, Healthcare Utilization, Occupational Exposure, Brazil

JEL Classification: I13 , I18 , C26

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Resumo

Este estudo fornece novas evidências causais sobre os efeitos do plano privado de saúde na utilização dos serviços de saúde no Brasil, considerando tanto a margem extensiva (procura por atendimento de saúde) quanto a margem intensiva (número de consultas médicas). Utilizando microdados da Pesquisa Nacional de Saúde (PNS) de 2019, exploramos variações exógenas na cobertura por plano de saúde entre grupos ocupacionais como variável instrumental para o tratamento endógeno de possuir plano privado de saúde. Nossa estratégia de identificação baseia-se em diferenças na difusão dos planos de saúde entre grupos definidos pela atividade econômica, formalização do emprego e estado de residência, capturando mecanismos institucionais e relacionados ao mercado de trabalho que afetam o acesso ao seguro privado de saúde. Controlando por características demográficas e socioeconômicas, condições crônicas de saúde e fatores regionais, os resultados indicam que o plano privado de saúde aumenta substancialmente a utilização dos serviços de saúde entre indivíduos ocupados. Na margem intensiva, as estimativas sugerem que possuir plano privado de saúde mais do que dobra o número anual de consultas médicas, com resultados robustos a especificações alternativas, transformações da variável dependente e diferentes desenhos amostrais. Os resultados também revelam importantes efeitos heterogêneos. O impacto do plano privado de saúde sobre as consultas médicas é significativamente maior entre homens, indivíduos sem doenças crônicas e indivíduos mais jovens. Em particular, o efeito marginal do plano de saúde diminui com a idade, sugerindo que a cobertura privada gera maiores ganhos de utilização entre grupos com menor nível inicial de uso dos serviços de saúde. Em relação à margem extensiva, o plano privado de saúde aumenta moderadamente a probabilidade de procurar atendimento médico, mas eleva substancialmente a intensidade da utilização dos serviços de saúde condicional ao acesso.

Palavras-chave: Plano de Saúde, Utilização dos Serviços de Saúde, Exposição Ocupacional, Brasil.

1 Introduction

The private health insurance and health plan market is characterized by uncertainty about the occurrence and costs of illnesses experienced by economic agents [1]. This environment of uncertainty gives rise to market failures associated with information asymmetry, such as adverse selection [2] and moral hazard [3–5]. Based on this literature, adverse selection arises from pre-contractual information asymmetry between individuals and health insurers or health plan providers. In this context, individuals possess more information about their expected risk of illness than insurers do. Consequently, those with a higher probability of becoming ill or a greater expected need for healthcare services have stronger incentives to purchase private health insurance or enroll in private health plans. In contrast, moral hazard occurs after insurance coverage has been obtained. Individuals may alter their healthcare utilization behavior due to reduced marginal out-of-pocket costs, which can increase demand for medical services.

These characteristics of the private health insurance market inevitably impose important challenges for identifying the causal effects of health insurance coverage on healthcare utilization, given the endogenous nature of the treatment variable (having or not having health insurance coverage). First, individuals at higher risk of illness or with a greater expected need for healthcare services are also more likely to obtain health insurance coverage, leading to a non-random distribution of treatment. Moreover, having health insurance is strongly associated with socioeconomic status, income, and education, as well as labor market conditions, such as access to employer-sponsored health plans. As a result, individuals with and without health insurance may systematically differ in both observable and unobservable characteristics that also directly affect healthcare utilization. Therefore, a simple comparison of healthcare utilization between insured and uninsured individuals would yield a biased estimate of this relationship.

Accordingly, the health economics literature has increasingly focused on investigating the causal effects of health insurance coverage on healthcare utilization. The study conducted by Joseph P. Newhouse [6], for instance, is among the most influential pieces of evidence on the causal impact of health insurance. Using data from individuals randomly assigned to health plans with varying copayment and coverage in the RAND Health Insurance Experiment (HIE), the study overcame endogeneity associated with the treatment variable. The growing use of quasi-experimental methods to identify the positive causal effects of health insurance coverage on healthcare utilization should also be highlighted [7–13]. Nevertheless, studies estimating this causal relationship in developing countries such as Brazil remain relatively scarce [14, 15]. In general, most existing studies rely on observational comparisons between insured and uninsured individuals, disregarding the endogenous nature of health insurance coverage in Brazil [16, 17].

In this context, a noteworthy feature of the Brazilian healthcare system is the coexistence of public and private healthcare services. To ensure universal, comprehensive, and equitable access to healthcare throughout the country, the Brazilian Unified Health System, known as the Sistema Único de Saúde (SUS), was established through Law No. 8,080 of September 19, 1990. Furthermore, the 1988 Brazilian Federal Constitution defines private healthcare provision as complementary to the public healthcare system, although the sector was only formally regulated by Law No. 9,656 of 1998. This regulatory framework consolidated Brazil’s supplementary healthcare market and enabled beneficiaries to simultaneously access both private healthcare services and the public services provided by the SUS.

Currently, Brazil’s private health insurance market is regulated by the National Supplementary Health Agency (ANS) and has experienced substantial growth in recent years. According to data published by the ANS, more than 53 million beneficiaries were enrolled in medical assistance plans in December 2025, representing a 10.9% increase over December 2020 [18]. However, even before the creation of the SUS, the private health insurance market had already demonstrated strong growth potential in Brazil. A substantial rise in the number of workers enrolled in private health plans was observed due to the reduced financial risk associated with illness. This trend continued after the establishment of the SUS, largely due to the limited availability and relatively low quality of public healthcare services at that time [19].

Another important characteristic of Brazil’s private health insurance market is that most supplementary healthcare coverage stems from individuals’ participation in the labor market, particularly through employer-sponsored health plans. Bahia [20] argues that the Brazilian health insurance market is predominantly structured around occupational and institutional ties rather than purely individual consumer choices. Consequently, access to private health insurance in Brazil occurs primarily through employment relationships and corporatist arrangements, rather than through autonomous household decisions in the marketplace. As a result, the expansion of supplementary healthcare coverage in Brazil has historically followed the dynamics of formal labor market growth. Additionally, the supplementary healthcare market in Brazil exhibits substantial heterogeneity in health insurance coverage across occupational groups.

Workers in certain sectors and occupations display significantly higher rates of private health insurance coverage than others, reflecting institutional differences within the labor market.

Although previous studies, such as Meer and Rosen [21], have examined the causal relationship between health insurance coverage and healthcare utilization using instrumental variable strategies, our study contributes to the literature by providing new causal evidence on the effects of private health insurance on healthcare utilization in Brazil. Using data from the 2019 National Health Survey (PNS), we exploit a potential source of exogenous variation in health insurance coverage across occupational groups as an instrument for private health insurance status to identify the causal effect on healthcare utilization. This strategy allows us to address endogeneity concerns arising from adverse selection and other sources of non-random insurance coverage. We examine both the extensive margin of healthcare utilization, captured by a dummy variable indicating healthcare-seeking behavior over the previous two weeks, and the intensive margin, measured by the annual number of medical consultations. In addition, we investigate potential heterogeneous effects on healthcare utilization according to sex, age, and the presence of chronic diseases.

The main findings of our study indicate that private health insurance substantially increases healthcare utilization among employed individuals. The estimates suggest that, after addressing endogeneity through an instrumental variable approach, having private health insurance more than doubles the annual number of medical consultations, revealing a downward bias in the Ordinary Least Squares (OLS) estimates. These results remain robust across alternative model specifications, including different transformations of the dependent variable and alternative instrumental variable constructions that do not incorporate survey weights. Our findings also reveal significant heterogeneity in the causal effects of private health insurance across gender, age, and health status. The impact of private health insurance on medical consultations is significantly larger among men, individuals without chronic diseases, and younger individuals. In particular, the marginal effect of health insurance declines with age, suggesting that private coverage generates stronger utilization gains among groups with lower baseline levels of healthcare use. Regarding the extensive margin, private health insurance moderately increases the probability of seeking medical care, while exerting a much stronger effect on the intensive margin by substantially increasing the frequency of healthcare utilization conditional on access.

Beyond this introduction, the remainder of this paper is organized into six sections. Section 2 provides an overview of the private health insurance market in Brazil. Section 3 reviews the existing empirical literature on the causal effects of health insurance on healthcare utilization, highlighting methodological approaches and key findings from both developed and developing countries. Section 4 describes the data source, variable definitions, and descriptive statistics. Section 5 presents the empirical strategy adopted to identify the causal effects of private health insurance on healthcare utilization. Section 6 reports and discusses the main results. Finally, Section 7 concludes and outlines the main implications of the findings.

2 The supplementary healthcare market in Brazil

The Brazilian health system is organized around a dual structure in which a universal public system coexists with a supplementary private sector. The *Sistema Único de Saúde* (SUS), established by the 1988 Federal Constitution, guarantees universal and free access to health services for all Brazilian citizens [19, 22]. However, despite the constitutional mandate, persistent underfunding, long waiting times, and geographical imbalances in the supply of health services have historically led a substantial share of the population to seek care in the private sector [14, 17]. This arrangement characterizes a “duplicative” model of private health insurance, in which the supplementary sector covers services that are also nominally available through the public system, rather than filling gaps left by public provision [19, 23].

The legal and regulatory framework for the supplementary health market was established by Law 9,656 of 1998, which, for the first time, imposed a comprehensive set of rules governing contracting, coverage, and operations of private health plans in Brazil [19]. Prior to this legislation, the market operated with minimal oversight, generating widespread consumer protection problems, including the arbitrary cancellation of contracts and the denial of coverage for high-cost procedures [14, 19]. The creation of the *Agência Nacional de Saúde Suplementar* (ANS) in 2000 consolidated the regulatory framework, attributing to a dedicated agency the responsibilities of market oversight, product regulation, and consumer protection [18, 19]. A key provision of Law 9,656 requires private health plan operators to reimburse the SUS for services delivered to their beneficiaries by public facilities, establishing a formal financial link between the two systems [19, 23].

The supplementary health sector has experienced substantial growth since its regulatory consolidation. At the end of 2025, the ANS reported approximately 53.3 million beneficiaries enrolled in medical-hospital plans and 35.1 million in dental plans, representing a steady year-over-year expansion, driven in part by the recovery of formal employment and the growing penetration of employer-sponsored group plans [18]. Coverage, however, remains highly concentrated among higher-income and formally employed segments of the population [14, 16]. Evidence from successive waves of nationally representative household surveys indicates that private health insurance coverage rates among individuals in the richest income quintile are an order of magnitude higher than those observed in the poorest quintile, a pattern that has remained largely stable despite the overall expansion of coverage [14, 17]. These disparities are consistent with theoretical predictions of insurance market equilibrium models under imperfect information [1, 2].

In addition to income-based disparities, there are pronounced regional differences in supplementary health coverage across Brazilian states. Data from the ANS and nationally representative surveys indicate that coverage rates range from below 10% of the population in the states of the North and Northeast regions to above 40% in São Paulo and other states of the Southeast region [18, 24]. These disparities reflect both differences in economic development and labor market formalization across regions and the historically unequal distribution of private healthcare infrastructure [19, 24]. The regional heterogeneity in health insurance coverage is not merely a reflection of income differences; it also reflects variation in the availability and organization of supplementary healthcare networks, including the presence of large health maintenance organizations and employer-based group plans, which are concentrated in more industrialized regions [23, 25].

A distinctive feature of the Brazilian supplementary market is its close linkage to the formal labor market through employer-sponsored group health plans. Brazilian firms, particularly in the formal sector of the economy, routinely offer private health insurance as an employment benefit, a practice that reflects both the legal framework governing labor relations and the role of health benefits as a tool for attracting and retaining workers [25, 26]. The concentration of health insurance coverage among formally employed workers is particularly pronounced: among workers in the formal labor market, roughly one-third hold employer-sponsored plans, whereas coverage rates among informal workers are substantially lower [14, 18]. This institutional structure generates substantial heterogeneity in access to private health insurance across occupational groups and employment arrangements, which has important implications for the identification of causal effects of insurance coverage on healthcare utilization [15, 16].

The dual structure of the Brazilian health system raises important concerns about horizontal equity in healthcare utilization. Studies using data from successive waves of the National Household Sample Survey (PNAD) and the National Health Survey (PNS) consistently document that individuals with private health insurance exhibit higher rates of physician consultations, preventive care use, and specialist visits than uninsured individuals, even after controlling for health status and socioeconomic characteristics [14, 16, 17]. However, the direction of causality in this relationship is not straightforward, as individuals with higher expected healthcare needs may be more likely to acquire private health insurance in the first place [1, 2]. Disentangling the causal effect of insurance coverage from these selection processes requires rigorous identification strategies, motivating the empirical approach adopted in the present study.

3 Previous evidence regarding the causal effect of private health insurance on healthcare utilization

The theoretical foundations for analyzing the relationship between health insurance and healthcare utilization are rooted in the economics of information and incentives. Seminal contributions by Arrow [1] established that health insurance markets are characterized by pervasive information asymmetries, generating welfare relevant departures from competitive equilibrium. Building on this framework, Rothschild and Stiglitz [2] demonstrated that adverse selection, whereby individuals with higher expected healthcare needs are more likely to seek insurance coverage, can destabilize competitive insurance markets. From the perspective of healthcare utilization, Pauly [3] introduced the concept of moral hazard, arguing that insurance reduces the marginal cost of care at the point of consumption, thereby inducing insured individuals to utilize more healthcare services than they would in the absence of coverage. Subsequent theoretical work further refined the welfare implications of moral hazard. In particular, Nyman [5] distinguished between inefficient overconsumption induced by lower out of pocket prices and welfare improving income transfer effects that arise when insurance enables sick individuals to access healthcare services that would otherwise be unaffordable. Related contributions by Arnott and Stiglitz [27] emphasized the broader welfare distortions generated by asymmetric information and moral hazard in insurance markets.

The RAND Health Insurance Experiment (HIE), conducted between 1974 and 1982, represents the most influential empirical evidence on the causal relationship between health insurance generosity and healthcare utilization [6]. By randomly assigning approximately 2,000 non-elderly families in the United States to insurance plans with varying levels of cost-sharing, the RAND HIE generated experimental estimates of the price elasticity of demand for medical care. The central finding was that healthcare utilization decreases significantly as cost-sharing increases, confirming the existence of moral hazard [6, 28]. Crucially, however, the experiment also documented that reductions in utilization induced by cost-sharing were not limited to discretionary or low-value care; they also encompassed the use of services for which medical benefit was demonstrated, raising concerns that cost-sharing may have adverse health consequences, particularly for low-income individuals [6, 29].

A second landmark contribution to the causal literature is the Oregon Health Insurance Experiment, which exploited a lottery based expansion of the Medicaid program in 2008 to estimate the effect of gaining public health insurance on healthcare utilization [8, 9]. Lottery selection served as an instrument for Medicaid enrollment, allowing the identification of credible causal effects under random assignment. The first year results reported by Finkelstein et al. [8] showed that lottery winners, who had a higher probability of Medicaid enrollment, experienced substantially greater utilization of primary and preventive healthcare services, lower out of pocket expenditures, and improvements in self reported health relative to non selected individuals. Subsequent evidence based on administrative records showed that Medicaid coverage also increased emergency department utilization, a finding consistent with previously unmet healthcare needs [30]. Additional studies using data from community health centers and other healthcare settings reinforced the conclusion that insurance coverage substantially increases the utilization of ambulatory healthcare services [10, 31].

Quasi-experimental evidence from the expansion of public and private insurance programs in the United States has further extended our understanding of the causal effects of insurance on healthcare utilization. Card et al. [7] used the sharp discontinuity in Medicare eligibility at age 65 as an instrument to estimate the effect of near-universal insurance coverage on hospital utilization among older Americans, finding substantial increases in inpatient care and physician visits at the eligibility threshold. Studies examining the Medicaid expansions under the Affordable Care Act (ACA) have documented significant increases in outpatient visits, preventive screenings, and substance abuse treatment among the newly insured [12, 13]. Importantly, several of these studies find that the insurance-induced increases in utilization are larger among previously uninsured individuals with greater unmet healthcare needs, suggesting that the effects are partly driven by access to care that individuals could not afford prior to gaining coverage [5, 12].

The international evidence from middle-income countries, while less extensive than the literature from the United States, also documents positive causal effects of insurance coverage on healthcare utilization. Studies from Colombia, Mexico, and other Latin American countries have used quasi-experimental designs to show that expansions in health insurance coverage lead to increased use of preventive and curative care, particularly among low-income populations [23]. In the Brazilian context, evidence on the causal effects of private health insurance is more limited. Andrade and Maia [14] documented large differences in healthcare utilization between insured and uninsured individuals in the supplementary sector, but acknowledged that these estimates are likely contaminated by selection bias. Politi [16] provided descriptive evidence on the concentration of healthcare demand among higher-income and insured segments of the population, documenting persistent socioeconomic gradients in the use of physician consultations and preventive services. Chiavegatto Filho et al. [17] used multilevel analysis to identify individual and contextual determinants of healthcare utilization in the metropolitan region of São Paulo, finding that private health insurance remains one of the strongest predictors of use even after controlling for health status and socioeconomic characteristics.

The study most closely related to the present paper in the Brazilian context is that of Menezes-Filho and Politi [15], who employed a regression kink design to estimate the causal effect of private health insurance on healthcare utilization. By exploiting discontinuous changes in the price of health insurance plans at income tax thresholds, where the value of tax deductions for health expenditures changes, the authors found that the induced variation in health insurance coverage increases the utilization of physician visits and preventive services such as mammography and Pap smear exams among women, while producing no significant effects on inpatient care or surgical procedures. This evidence supports the hypothesis that private health insurance expands both the extensive and intensive margins of ambulatory healthcare utilization, particularly for services commonly provided in the supplementary healthcare sector

and subject to substantial access barriers within the public healthcare system [14, 15]. The present study builds on and extends this literature by employing an instrumental variables strategy based on occupational and regional variation in health insurance coverage, which exploits a distinct source of exogenous variation and enables estimation of the causal effect of insurance across the full distribution of employed workers in Brazil.

4 Data

This study uses microdata from the 2019 National Health Survey (PNS), conducted by the Brazilian Institute of Geography and Statistics (IBGE). According to the Instituto Brasileiro de Geografia e Estatística [32], the PNS is a nationwide household survey designed to collect detailed information on the health conditions of the Brazilian population. In addition to socioeconomic and demographic characteristics of household residents, the 2019 edition provides extensive information on health-related topics, including access to and utilization of healthcare services, aspects of the Family Health Strategy, visits by endemic disease control agents, private health insurance coverage, medical care, self-perceived health status, lifestyle habits, chronic diseases, and other specific health-related issues.

The PNS covers the entire Brazilian territory, based on Census Tracts from the Geographic Operational Base, excluding areas with special characteristics as defined by the IBGE. The survey adopts a three-stage cluster sampling design, with stratification of the Primary Sampling Units. In the 2019 edition of the PNS, 108,525 households were selected [32], which comprises a final sample of 293,726 individuals. After excluding individuals outside the labor force and observations with missing information on the selected variables and after applying sampling weights, the final analytical sample comprises 50,514 employed individuals.

In this study, the treatment variable corresponds to a binary indicator of whether the individual is covered by private health insurance. The causal effect of private health insurance coverage on healthcare utilization, measured primarily by the number of medical consultations in the previous 12 months, captures the intensive margin of healthcare utilization. In addition, the impact on the extensive margin of healthcare utilization was estimated, measured by a binary indicator of whether the individual sought healthcare services during the two weeks preceding the reference date of the National Health Survey (PNS). Table A1, in Appendix A, presents the explanatory variables relevant to healthcare utilization that were included as control variables in the estimations.

The choropleth maps presented in Figure 1 display the weighted distribution of the two health care utilization measures across Brazilian states. Regarding the average of $\ln(\text{Medical Consultations} + 1)$, a clear geographic gradient emerges, with the highest values concentrated in the South and Southeast regions, particularly in São Paulo, Rio de Janeiro, Minas Gerais, and the southern states, where the mean log consultation rate exceeds 1.1. In contrast, states in the North and parts of the Northeast, such as Amazonas, Pará, and Maranhão, exhibit the lowest values, falling below 0.9. This pattern is consistent with the well-documented disparities in healthcare infrastructure, physician density, and private health insurance penetration across Brazilian macro-regions.

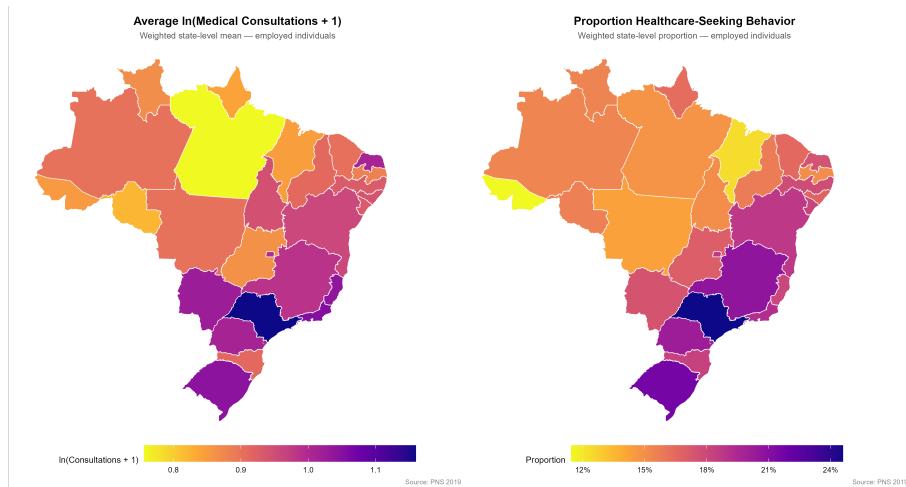


Fig. 1: Geographic Distribution of Healthcare Utilization Across Brazilian States

The distribution of health care-seeking behavior follows a broadly similar geographic pattern, with the highest proportions, above 21%, concentrated in the South and Southeast, and the lowest, below 15%, observed in the North and parts of the Center West. Notably, the Distrito Federal stands out as an exception in the Center-West, displaying relatively high utilization in both measures, likely reflecting its atypical socioeconomic profile and concentration of public-sector employment with employer-sponsored health benefits. Taken together, both maps underscore the persistent North-South divide in healthcare utilization in Brazil, reinforcing the importance of controlling for state fixed effects in the empirical strategy to account for unobserved regional heterogeneity. Additional descriptive statistics are reported in Table 1.

Table 1: Descriptive Statistics

Variable	Obs.	Mean	Std. Dev.	Min	Max
Medical Consultations	50,514	2.687	3.925	0	250
Private Health Insurance	50,514	0.304	0.460	0	1
Female	50,514	0.447	0.497	0	1
White	50,514	0.444	0.497	0	1
Age	50,514	40.403	13.497	15	98
No Schooling	50,514	0.027	0.161	0	1
Incomplete Primary Education	50,514	0.227	0.419	0	1
Complete Primary Education	50,514	0.081	0.272	0	1
Incomplete Secondary Education	50,514	0.070	0.256	0	1
Complete Secondary Education	50,514	0.336	0.472	0	1
Incomplete Higher Education	50,514	0.056	0.230	0	1
Complete Higher Education	50,514	0.204	0.403	0	1
Per Capita Household Income	50,514	1830.541	3066.349	13	203196
Hypertension	50,514	0.177	0.382	0	1
Diabetes	50,514	0.049	0.216	0	1
High Cholesterol	50,514	0.117	0.321	0	1
Heart Disease	50,514	0.031	0.174	0	1
Stroke	50,514	0.008	0.087	0	1
Asthma	50,514	0.047	0.211	0	1
Arthritis	50,514	0.047	0.212	0	1
Chronic Back Pain	50,514	0.028	0.165	0	1
Depression	50,514	0.083	0.276	0	1
Pulmonary Disease	50,514	0.011	0.105	0	1
Cancer	50,514	0.016	0.127	0	1
Kidney Disease	50,514	0.011	0.104	0	1
Urban	50,514	0.876	0.330	0	1

Source: Authors' elaboration based on data from the 2019 National Health Survey (PNS).

5 Methodology

5.1 Identification strategy

The main objective of this study is to provide new evidence on the causal effect of private health insurance coverage on healthcare utilization in Brazil. A fundamental challenge in this type of analysis is the absence of the counterfactual outcome, that is, the level of healthcare utilization that insured individuals would have exhibited had they not been covered by private health insurance. Since the counterfactual outcome is unobservable, the causal effect of private health insurance cannot be directly identified at the individual level.

In experimental settings, random assignment of the treatment ensures that individuals in both groups are statistically comparable. Thus, a simple comparison of observed outcomes between the two groups provides a causal estimate of the Average Treatment Effect (ATE), since the only difference between the two groups is private health insurance coverage. However, in the Brazilian context, health insurance coverage is not randomly assigned. In general, the endogenous nature of private health insurance coverage arises from several factors, including self-selection based on expected healthcare needs (adverse selection), socioeconomic characteristics, and labor market conditions. Consequently, insured and uninsured individuals may differ systematically in both observable and unobservable characteristics that also influence healthcare utilization. Thus, coefficients obtained from standard Ordinary Least Squares (OLS) regressions based on simple comparisons between individuals with and without private health insurance are likely to produce biased estimates of the causal effect of insurance coverage on healthcare utilization. Even after controlling for a rich set of observable characteristics, unobserved factors related to health risks, preferences for medical care, and labor market attachment may continue to generate endogeneity concerns. Therefore, identifying exogenous variation in health insurance coverage is crucial for obtaining credible causal estimates.

Given the observational nature of the PNS data and the endogenous nature of private health insurance coverage, this study employs an instrumental variables (IV) approach to identify the causal effect of health insurance on healthcare utilization. The instrument exploits heterogeneity in private health insurance coverage rates across occupational groups, differences in the degree of employment formalization, and regional variation in health insurance coverage. Thus, the identifying variation is derived from occupational differences in access to employer-sponsored health insurance plans, combined with labor market formalization and regional coverage patterns, as specified in the following equation:

$$Z_i = \frac{\sum_{j \in c, j \neq i} w_j \text{Plano}_j}{\sum_{j \in c, j \neq i} w_j} \quad (1)$$

The instrument Z_i is defined as the leave-one-out weighted health insurance coverage rate within cells formed by occupational sector, employment formalization status, and state of residence. Excluding the individual from the calculation prevents mechanical correlation between the instrument and the individual's insurance status. Thus, in this study, Z_i denotes the instrumental variable assigned to employed individual i ; Plan_j is an indicator variable equal to 1 if employed individual j has private health insurance coverage and 0 otherwise; w_j represents the survey weight associated with employed individual j ; c denotes the cell defined by the combination of occupational group, formalization status in employment, and state of residence; condition $j \neq i$ indicates that the individual i is excluded from the calculation of the cell-specific coverage rate (left-out construction). Since the analysis is restricted to employed individuals, all cell-specific coverage rates are computed exclusively within the employed population.

To be considered a valid instrument for the endogenous variable that indicates private health insurance coverage, the instrument must satisfy two fundamental conditions in the instrumental variables framework. First, the instrument must be relevant, meaning that it must be correlated with the endogenous treatment variable. Given the construction of Z_i in this study, instrument relevance is grounded in the institutional structure of the Brazilian private health insurance market, which is closely linked to labor market participation, particularly through employer-sponsored group health plans. Consequently, workers in occupational groups that have historically had higher rates of employer-provided health insurance are more likely to be covered by private health insurance at the individual level. A similar argument applies to employment formalization, since health insurance benefits are considerably more prevalent among formally employed workers. Finally, regional variation in health insurance coverage captures differences in the availability and penetration of private health insurance markets across states. States with higher coverage rates tend to offer greater opportunities to obtain private health insurance, reflecting differences in the regional organization and availability of supplementary healthcare services. As a result, individuals residing in those states are more likely to be covered by private health insurance.

Second, the instrument must be uncorrelated with the error term in the healthcare utilization equation and must not directly affect healthcare utilization. In other words, the exclusion restriction requires that the instrument Z_i affects individual healthcare utilization only through its effect on private health insurance coverage. Although this assumption cannot be tested directly, we argue that after controlling for demographic, socioeconomic, health, and geographic characteristics, the instrument captures variation in access to private health insurance attributable to occupational and institutional features of the labor market rather than to individual preferences for healthcare or underlying healthcare needs. Under this assumption, individuals exposed to occupational groups and regional environments with greater diffusion of private health insurance are more likely to obtain coverage due to occupational and institutional factors rather than because of differences in healthcare preferences or expected healthcare needs.

5.2 Empirical strategy

To estimate the causal effects of private health insurance on healthcare utilization, we employ an instrumental variable (IV) strategy based on exogenous variation in health insurance coverage across occupational groups. Using the Two-Stage Least Squares (2SLS) estimator, the first stage consists of estimating the following equation:

$$\text{Plan}_i = \alpha_0 + \alpha_1 Z_i + X_i' \gamma + v_i \quad (2)$$

where Plan_i is a binary indicator equal to one if individual i is covered by a private health insurance plan; Z_i represents the instrumental variable based on occupational health insurance coverage rates, satisfying the exclusion restriction; X_i is a vector of control variables, including a set of demographic characteristics (such as gender, race, and age), a set of socioeconomic characteristics (education dummies and per capita

income), a vector of 12 chronic disease indicators, and a vector including state and area of residence dummies; and v_i is the error term.

In the second stage, the causal effect of private health insurance coverage on healthcare utilization is obtained by estimating the following equation:

$$\ln(Utilization_i) = \beta_0 + \beta_1 \widehat{Plan}_i + X_i' \delta + u_i \quad (3)$$

where $Utilization_i$ is the outcome variable of interest in this study, corresponding to healthcare utilization by individual i . The coefficient β_1 captures the causal effect of private health insurance coverage on healthcare utilization. X_i' contains the same control variables included in Equation 2, and u_i is the error term of the structural equation. The causal effect estimated in Equation 3 is examined for both the intensive and extensive margins of healthcare utilization. In the first case, the intensive margin is measured by the number of medical consultations during the previous year. In contrast, the extensive margin is measured by a binary indicator of whether the individual sought healthcare services during the two weeks prior to the survey. The IV/2SLS estimator was implemented using the `ivregress` command in Stata, accounting for the complex survey design of the National Health Survey (PNS), including sampling weights, stratification, and primary sampling units.

6 Results and Discussion

6.1 Effects of Health Insurance on the Intensive Margin of Healthcare Utilization

The main objective of this study is to provide new evidence on the causal relationship between private health insurance coverage and healthcare utilization. As previously discussed, because private health insurance coverage is not randomly assigned across individuals, coefficients estimated through standard regression methods are likely to produce biased estimates of the effect of health insurance on healthcare utilization. This occurs due to adverse selection, as individuals possess more information about their own health risks and expected healthcare needs than insurers do. Consequently, individuals with greater expected healthcare needs are also more likely to purchase private health insurance and to use healthcare services more intensively. In addition, insured and uninsured individuals may differ systematically in several other observable and unobservable characteristics that also affect healthcare utilization. The results reported in Table 2 illustrate how systematic differences in observable characteristics may bias OLS estimates.

Table 2 reports seven different specifications estimated using Ordinary Least Squares (OLS). In the first column, we estimate the natural logarithm of the number of medical consultations (plus one) as a function solely of the treatment dummy variable indicating private health insurance coverage. In the subsequent specifications, we sequentially add four groups of control variables corresponding to demographic characteristics, socioeconomic characteristics, chronic health conditions, and, in the fifth specification, regional controls, including an urban residence dummy and state fixed effects; the coefficients for these controls are omitted from the table for brevity.

Across the first five specifications, the estimated effect of private health insurance on the average number of medical consultations remains positive and statistically significant. However, the estimated coefficient declines from 0.379 to 0.288 after the inclusion of the full set of controls. This substantial sensitivity of the estimated coefficient suggests selection on observable characteristics, indicating that part of the observed association between private health insurance coverage and the number of medical consultations is driven by differences in socioeconomic status, chronic health conditions, and demographic characteristics across individuals.

In the sixth specification, we examine whether the complex survey design influences the statistical significance of the estimated effects. The results indicate that the OLS estimator remains highly statistically significant even when the survey design is not accounted for, suggesting that the estimates from the fully specified model are robust to the inclusion of sampling weights and other survey design features.

Given that the survey design does not substantially alter the estimated conditional relationships, we estimate a seventh specification to further investigate whether systematic differences in observable characteristics affect the OLS estimates. For this purpose, we employ the entropy balancing approach proposed by Hainmueller [33], which reweights the treatment and control groups to achieve balance in observable covariates and their statistical moments. Appendix B, Table B2, shows that covariate balance was successfully achieved after applying entropy balancing. After the reweighting procedure, the same specification presented in Column 6 was re-estimated using OLS without explicitly incorporating the

survey sampling weights. The results indicate that the estimated effect in the entropy-balanced model remains relatively similar to the effects obtained in the previous specifications without entropy balancing (Columns 5 and 6). This finding suggests that the fully specified model already controls reasonably well for selection on observable characteristics.

Table 2: OLS Estimates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Plan	0.379*** (0.012)	0.355*** (0.012)	0.322*** (0.014)	0.298*** (0.014)	0.288*** (0.014)	0.289*** (0.008)	0.307*** (0.011)
Female		0.344*** (0.011)	0.341*** (0.011)	0.271*** (0.011)	0.267*** (0.011)	0.281*** (0.006)	0.253*** (0.012)
white		-0.014 (0.012)	-0.027* (0.012)	-0.031** (0.012)	-0.044*** (0.013)	-0.028*** (0.007)	-0.022+ (0.012)
Age		0.003 (0.003)	0.003 (0.003)	0.003 (0.002)	0.003 (0.002)	-0.003** (0.001)	-0.005+ (0.003)
Age Squared		0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	0.000* (0.000)	0.000 (0.000)
Incomplete Primary Educ.			0.031 (0.027)	0.004 (0.026)	-0.001 (0.025)	-0.027+ (0.016)	-0.129** (0.041)
Complete Primary Educ.			0.015 (0.033)	-0.014 (0.032)	-0.027 (0.032)	-0.046* (0.018)	-0.158*** (0.043)
Incomplete Secondary Educ.			0.025 (0.033)	0.002 (0.032)	-0.009 (0.032)	-0.033+ (0.019)	-0.142** (0.044)
Complete Secondary Educ.			0.041 (0.027)	0.031 (0.026)	0.012 (0.026)	-0.002 (0.016)	-0.108** (0.040)
Incomplete Higher Educ.			0.049 (0.036)	0.031 (0.036)	0.018 (0.036)	0.037+ (0.021)	-0.083+ (0.044)
Complete Higher Educ.			0.062* (0.030)	0.064* (0.030)	0.053+ (0.030)	0.060*** (0.018)	-0.048 (0.041)
Ln(PC Household Income)			0.025** (0.008)	0.021** (0.007)	0.011 (0.008)	0.013** (0.004)	0.021* (0.009)
hypertension				0.207*** (0.014)	0.201*** (0.014)	0.237*** (0.008)	0.191*** (0.015)
Diabetes				0.277*** (0.024)	0.272*** (0.024)	0.299*** (0.014)	0.256*** (0.026)
High Cholesterol				0.119*** (0.017)	0.118*** (0.017)	0.132*** (0.010)	0.132*** (0.017)
Heart Disease				0.160*** (0.033)	0.158*** (0.032)	0.181*** (0.017)	0.158*** (0.034)
Stroke				0.213** (0.080)	0.219** (0.078)	0.197*** (0.031)	0.095+ (0.056)
Asthma				0.132*** (0.030)	0.133*** (0.029)	0.114*** (0.015)	0.077** (0.027)
Arthritis				0.217*** (0.051)	0.217*** (0.048)	0.206*** (0.014)	0.224*** (0.025)
Chronic Back Pain				0.209*** (0.035)	0.186*** (0.034)	0.148*** (0.020)	0.158*** (0.039)
Depression				0.281*** (0.022)	0.280*** (0.022)	0.274*** (0.012)	0.311*** (0.023)
Pulmonary Disease				0.220*** (0.054)	0.209*** (0.053)	0.209*** (0.031)	0.212*** (0.046)
Cancer				0.283*** (0.054)	0.281*** (0.053)	0.325*** (0.024)	0.300*** (0.045)
Kidney Disease				0.215*** (0.051)	0.218*** (0.051)	0.196*** (0.029)	0.190** (0.070)
Urban					0.047*** (0.014)	0.031*** (0.008)	0.055** (0.017)
Intercept	0.891*** (0.007)	0.563*** (0.052)	0.376*** (0.073)	0.462*** (0.072)	0.412*** (0.075)	0.538*** (0.044)	0.556*** (0.096)
State Fixed Effects	No	No	No	No	Yes	Yes	Yes
R2	0.056	0.120	0.121	0.186	0.194	0.202	0.185
Observations	50514	50514	50514	50514	50514	50514	50514

Source: Authors' elaboration based on data from the 2019 National Health Survey (PNS). *Notes:* Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Coefficients for state dummy variables (UF fixed effects) are not reported due to space constraints. The omitted reference groups are male, non White, individuals with no formal education, individuals without chronic conditions, rural residents, and residents of the state of Rondônia.

However, even after controlling for important observable factors such as demographic characteristics, socioeconomic status, chronic health conditions, and regional aspects, the OLS estimates reported in

Table 2 are still likely to suffer from an important omitted variable bias affecting both the intensity of healthcare utilization and the decision to acquire private health insurance coverage, even after balancing the covariates through entropy balancing. Unobserved factors such as risk aversion, preventive behavior, preferences, and expectations regarding future healthcare needs are classic examples of omitted variables whose effects may be partially captured by the estimated coefficient of health insurance coverage, thereby rendering the OLS estimates biased and inconsistent for identifying the causal effect.

In this context, we employ an instrumental variable strategy that exploits variation in health insurance coverage across occupational groups, formal labor market attachment, and regional labor market structures, and estimate the model using Two-Stage Least Squares (2SLS). Given that access to private health insurance in Brazil is strongly associated with formal labor market participation, and that this association may be even stronger in certain occupational segments, it is reasonable to expect that workers employed in occupations and regions with higher rates of employer-sponsored health insurance coverage are also more likely to obtain private health insurance, independently of their individual preferences or healthcare utilization decisions. Thus, after controlling for demographic factors, socioeconomic characteristics, chronic health conditions, and regional aspects, differences in access to private health insurance between individuals employed in sectors with high rates of private health insurance coverage and those employed in sectors with lower coverage rates are likely to reflect institutional differences in the labor market, such as labor market segmentation and regional heterogeneity between formal and informal sectors, rather than individual preferences.

Proceeding with our empirical strategy, the main results (the estimates of the second-stage Equation 3) are reported in Table 3. Following the same reasoning adopted previously, we estimate the same five baseline specifications presented in Table 2, now using Two-Stage Least Squares (2SLS). As additional controls are sequentially introduced, the estimated effect of private health insurance on the number of medical consultations exhibits some sensitivity across specifications. Nevertheless, regardless of the specification considered, our findings consistently indicate that the effect of private health insurance remains positive and statistically significant at the 0.1% level, mirroring the pattern observed in the OLS estimates. Even in the fully specified model (Column 5), the IV estimate remains substantially larger than its OLS counterpart, suggesting that a substantial share of the OLS model’s bias arises from endogeneity rather than solely from selection on observable characteristics. Based on the fully specified model, the estimated causal effect of 0.711 implies that possessing private health insurance more than doubles the expected number of medical consultations, corresponding to an increase of $(100 \times (e^{0.711} - 1) = 103.6\%)$. This finding suggests that standard regression models substantially underestimate the effect of private health insurance on healthcare utilization, indicating a negative bias in OLS estimates. Consistent with Meer and Rosen [21], our results reinforce the concern that conventional estimators may be biased when health insurance coverage is endogenous to healthcare utilization decisions.

The negative bias may reflect factors such as unmet healthcare needs and other barriers to healthcare access that are not fully controlled for in the estimates, helping to explain its direction, particularly given that the correlation between private health insurance coverage and the instrument is positive. For uninsured individuals, for example, the greater the access constraints to healthcare services or the greater the unmet healthcare needs, the lower the observed healthcare utilization is likely to be. When estimating the effect of health insurance coverage on health indicators, Dor et al. [34] finds evidence of downward bias in OLS estimates relative to 2SLS instrumental-variables estimates, with the estimated effect up to six times larger than the OLS estimate.

The sixth specification aims to assess whether the complex survey design alters the results obtained from the fully specified model. This estimation was weighted only by the sampling weights and computed with robust standard errors. The results indicate that accounting for the survey design produces estimates that are nearly identical to those reported in Column 5 (Table 3).

6.2 Relevance of the Proposed Instrument

Having analyzed the causal effect, it remains to demonstrate that the instrument proposed in this study is indeed relevant. For the instrument to satisfy the first requirement for a valid instrumental variable candidate [35], it must be correlated with the endogenous regressor. We argue that employed individuals in certain occupational groups are more likely to have private health insurance. To assess this condition, Table 4 reports the first-stage equation estimates. Following the same specification strategy adopted previously, the first six columns correspond exactly to the first-stage equations associated with the models presented in Table 3, estimated using 2SLS. Additionally, as a robustness check, Column 7 reports the estimates from a logit regression model presented as average marginal effects. Overall, the results show

that all specifications support the hypothesis developed in this study, indicating that the instrument is strongly associated with private health insurance coverage and therefore satisfies the relevance condition required for a valid instrumental variable. The fully specified first-stage model (Column 5 of Table 4) reports a large t-statistic ($t=20.58$), which is statistically significant at the 1% level.

Table 3: Causal Effect of Health Insurance on Medical Consultations in the Last 12 Months

	(1)	(2)	(3)	(4)	(5)	(6)
Plan	0.629*** (0.032)	0.649*** (0.034)	0.933*** (0.084)	0.845*** (0.080)	0.711*** (0.081)	0.711*** (0.080)
Female		0.334*** (0.011)	0.341*** (0.012)	0.275*** (0.012)	0.271*** (0.011)	0.271*** (0.012)
White		-0.064*** (0.013)	-0.049*** (0.013)	-0.050*** (0.013)	-0.055*** (0.013)	-0.055*** (0.013)
Age		0.000 (0.003)	-0.001 (0.003)	-0.001 (0.003)	-0.000 (0.003)	-0.000 (0.003)
Age Squared		0.000* (0.000)	0.000* (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Incomplete Primary Educ.			0.024 (0.028)	-0.001 (0.027)	0.001 (0.026)	0.001 (0.027)
Complete Primary Educ.			-0.024 (0.034)	-0.048 (0.033)	-0.044 (0.032)	-0.044 (0.033)
Incomplete Secondary Educ.			-0.013 (0.035)	-0.032 (0.034)	-0.026 (0.033)	-0.026 (0.033)
Complete Secondary Educ.			-0.043 (0.030)	-0.043 (0.029)	-0.036 (0.028)	-0.036 (0.029)
Incomplete Higher Educ.			-0.102* (0.043)	-0.102* (0.042)	-0.077+ (0.040)	-0.077+ (0.041)
Complete Higher Educ.			-0.161*** (0.043)	-0.135** (0.042)	-0.094* (0.041)	-0.094* (0.041)
Ln(PC Household Income)			-0.069*** (0.015)	-0.063*** (0.015)	-0.050*** (0.015)	-0.050*** (0.014)
hypertension				0.201*** (0.014)	0.198*** (0.014)	0.198*** (0.015)
Diabetes				0.271*** (0.026)	0.269*** (0.025)	0.269*** (0.025)
High Cholesterol				0.099*** (0.019)	0.103*** (0.018)	0.103*** (0.018)
Heart Disease				0.137*** (0.035)	0.142*** (0.034)	0.142*** (0.034)
Stroke				0.237** (0.086)	0.236** (0.083)	0.236** (0.083)
Asthma				0.107*** (0.032)	0.115*** (0.031)	0.115*** (0.030)
Arthritis				0.218*** (0.054)	0.217*** (0.051)	0.217*** (0.051)
Chronic Back Pain				0.179*** (0.035)	0.167*** (0.034)	0.167*** (0.036)
Depression				0.269*** (0.024)	0.272*** (0.023)	0.272*** (0.022)
Pulmonary Disease				0.205*** (0.061)	0.202*** (0.058)	0.202*** (0.059)
Cancer				0.265*** (0.059)	0.268*** (0.057)	0.268*** (0.058)
Kidney Disease				0.221*** (0.049)	0.220*** (0.049)	0.220*** (0.049)
Urban					0.031* (0.015)	0.031* (0.014)
Intercept	0.815*** (0.010)	0.577*** (0.053)	1.044*** (0.120)	1.050*** (0.115)	0.891*** (0.122)	0.891*** (0.122)
State Fixed Effects	No	No	No	No	Yes	Yes
Observations	50514	50514	50514	50514	50514	50514

Source: Authors' elaboration based on data from the 2019 National Health Survey (PNS). *Notes:* Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Coefficients for state dummy variables (UF fixed effects) are not reported due to space constraints. The omitted reference groups are male, non White, individuals with no formal education, individuals without chronic conditions, rural residents, and residents of the state of Rondônia.

The estimated coefficients from both the linear probability model and the marginal effects from the logit specification indicate that individuals in occupational groups with higher health insurance coverage rates are more likely to have private health insurance. The result is consistent with the institutional structure of the Brazilian labor market, in which employers in certain economic sectors provide private health insurance to employees as a mechanism to reduce costs associated with workers' health risks and absenteeism. The relevance of the proposed instrument is consistent with the literature on employer-sponsored health insurance. Previous studies, such as Buchmueller and Monheit [25] and Mulligan [26], argue that employer-sponsored health insurance also serves as a mechanism by which firms align their incentives with the objective of maintaining a healthier and more productive workforce, while also helping to reduce adverse selection problems.

Table 4: First-Stage Estimates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Z	0.937*** (0.020)	0.893*** (0.020)	0.475*** (0.022)	0.472*** (0.022)	0.473*** (0.023)	0.473*** (0.022)	0.406*** (0.019)
Female		0.023** (0.007)	0.002 (0.007)	-0.003 (0.007)	-0.003 (0.007)	-0.003 (0.007)	0.002 (0.007)
White		0.093*** (0.008)	0.023** (0.007)	0.022** (0.007)	0.029*** (0.008)	0.029*** (0.008)	-0.019* (0.007)
Age		0.006*** (0.001)	0.004*** (0.001)	0.004** (0.001)	0.004** (0.001)	0.004** (0.001)	0.004** (0.001)
Age Squared		-0.000* (0.000)	-0.000* (0.000)	-0.000* (0.000)	-0.000** (0.000)	-0.000** (0.000)	-0.000* (0.000)
Incomplete Primary Educ.			-0.006 (0.010)	-0.007 (0.010)	-0.005 (0.010)	-0.005 (0.010)	0.033 (0.024)
Complete Primary Educ.			0.029* (0.014)	0.027+ (0.014)	0.030* (0.014)	0.030* (0.014)	0.095*** (0.026)
Incomplete Secondary Educ.			0.020 (0.015)	0.018 (0.015)	0.020 (0.015)	0.020 (0.015)	0.085** (0.027)
Complete Secondary Educ.			0.070*** (0.011)	0.069*** (0.011)	0.069*** (0.011)	0.069*** (0.011)	0.136*** (0.023)
Incomplete Higher Educ.			0.165*** (0.020)	0.163*** (0.020)	0.163*** (0.020)	0.163*** (0.019)	0.184*** (0.026)
Complete Higher Educ.			0.261*** (0.014)	0.260*** (0.014)	0.258*** (0.014)	0.258*** (0.015)	0.223*** (0.024)
Ln(PC Household Income)			0.127*** (0.004)	0.126*** (0.004)	0.131*** (0.004)	0.131*** (0.004)	0.141*** (0.005)
hypertension				0.004 (0.009)	0.005 (0.009)	0.005 (0.009)	0.005 (0.009)
Diabetes				0.002 (0.014)	0.003 (0.014)	0.003 (0.014)	0.002 (0.013)
High Cholesterol				0.039*** (0.011)	0.037*** (0.011)	0.037*** (0.011)	0.032** (0.010)
Heart Disease				0.043* (0.020)	0.043* (0.020)	0.043* (0.019)	0.043* (0.018)
Stroke				-0.040 (0.031)	-0.039 (0.031)	-0.039 (0.030)	-0.035 (0.038)
Asthma				0.042** (0.016)	0.040* (0.016)	0.040* (0.016)	0.036* (0.014)
Arthritis				0.004 (0.018)	0.003 (0.018)	0.003 (0.017)	0.004 (0.017)
Chronic Back Pain				0.038 (0.025)	0.038 (0.025)	0.038 (0.025)	0.039+ (0.021)
Depression				0.016 (0.013)	0.018 (0.013)	0.018 (0.013)	0.010 (0.012)
Pulmonary Disease				0.009 (0.034)	0.011 (0.034)	0.011 (0.034)	0.005 (0.031)
Cancer				0.035 (0.027)	0.035 (0.027)	0.035 (0.028)	0.025 (0.024)
Kidney Disease				-0.012 (0.029)	-0.005 (0.029)	-0.005 (0.028)	0.001 (0.027)
Urban					0.005 (0.007)	0.005 (0.006)	0.074*** (0.011)
Intercept	0.019*** (0.005)	-0.180*** (0.028)	-0.944*** (0.038)	-0.930*** (0.038)	-1.019*** (0.040)	-1.019*** (0.040)	
State Fixed Effects	No	No	No	No	Yes	Yes	Yes
F Statistic	2236.011	538.330	775.988	409.401	228.562		77.51
Observations	50514	50514	50514	50514	50514	50514	50514

Source: Authors' elaboration based on data from the 2019 National Health Survey (PNS). Notes: Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Coefficients for state dummy variables (UF fixed effects) are not reported due to space constraints. The omitted reference groups are male, non-White individuals with no formal education, individuals without chronic conditions, rural residents, and residents of the state of Rondônia.

Consequently, individuals employed in specific formal-sector occupations and located in particular geographic regions are substantially more likely to possess private health insurance coverage than comparable individuals in other labor market segments. The relationship between occupational groups and private health insurance coverage is consistent with the organization of employer-sponsored insurance markets discussed by Folland et al. [23].

However, a concern raised by Bound et al. (1995) [36] is that 2SLS estimates may be problematic when instruments are exogenous but weak, potentially yielding misleading results even in large samples. Keane and Neal (2023, 2024) [37, 38] further argue that, in the presence of weak instruments, the conventional t-statistic may suffer from size distortions and power asymmetry, meaning that “2SLS standard errors tend to be artificially small when the 2SLS estimate is close to OLS and large when the 2SLS estimate is far from OLS.” When comparing the fully specified OLS estimate (Column 5 of Table 2) with the corresponding fully specified 2SLS estimate (Column 5 of Table 3), this issue does not appear to characterize our empirical results, since the IV estimate differs substantially from its OLS counterpart.

Even if the instrument were weak, Staiger and Stock (1997) [39] argue that, in exactly identified settings such as ours, the problems associated with inflated t-statistics under weak instruments are generally modest as long as the first-stage F-statistic exceeds 10. Keane and Neal (2023) [37], however, argue that the power asymmetry problem may persist even when the first-stage F-statistic is well above 10 or above the critical values proposed by Stock and Yogo (2005) [40]. In this context, the literature has increasingly relied on the Anderson–Rubin (AR) test (Anderson and Rubin, 1949) [41] to avoid the power asymmetry problem, with some authors even advocating the complete abandonment of the conventional t-test Keane and Neal (2024) [38].

To further assess instrument validity, we estimate the specification reported in Column 6 using the `ivreg2` command in Stata, as the `ivregress` command does not provide weak-identification diagnostics or weak-instrument-robust inference procedures. Because `ivreg2` does not allow the incorporation of the full complex survey design, the estimation is performed using sampling weights and heteroskedasticity-robust standard errors. Comparing the results with those reported in Column 5, the survey design appears to have little impact on the estimated coefficients. This finding increases our confidence in the diagnostic tests obtained from the `ivreg2` estimation.

Table ?? reports the Kleibergen–Paap rk LM test, the Cragg–Donald Wald F-statistic together with the Stock–Yogo critical values, and weak-instrument-robust inference based on the Anderson–Rubin test. The relevant F-statistics, 1908.87 for the Cragg–Donald test and 456.50 for the Kleibergen–Paap test, are substantially higher than the Stock-Yogo critical values, indicating that the instrument is strongly correlated with the endogenous variable.

In addition, the Kleibergen–Paap rk LM statistic rejects the null hypothesis of underidentification. Following the concerns raised by Keane and Neal (2023, 2024) regarding the potential size distortions and power asymmetry associated with conventional 2SLS t-tests, we additionally report the Anderson–Rubin (AR) weak-instrument-robust test. The null hypothesis that private health insurance has no causal effect on healthcare utilization is strongly rejected ($F = 86.03$, $p < 0.001$). This result provides additional evidence that the estimated 2SLS effect is not driven by weak-instrument problems or distortions associated with conventional t-tests in instrumental-variables models. Weak-instrument-robust inference based on the Anderson–Rubin test also rejects the null hypothesis that the endogenous regressor is jointly insignificant and that the orthogonality conditions are valid ($F=86.03$, $p<0.001$).

Table 5: Instrument Validity and Weak Instrument Tests

Statistic	Value	p-value
First-stage coefficient	0.473	0.000
Kleibergen–Paap rk LM statistic	420.4	0.000
Cragg–Donald Wald F statistic	1908.87	–
Kleibergen–Paap rk Wald F statistic	456.5	–
Stock–Yogo 10% critical value	16.38	–
Anderson–Rubin Wald test	86.03	0.000

Source: Authors’ elaboration based on data from the 2019 National Health Survey (PNS).

6.3 Robustness Checks

In the previous sections, we showed that the effect of private health insurance on the intensive margin of healthcare utilization is sensitive to the omission of important observable characteristics, as evidenced

by the variation in OLS and 2SLS coefficients as additional control variables were incorporated into the estimations. However, the estimated effects proved to be relatively stable with respect to both the exclusion of the survey design in the 2SLS estimations and the incorporation of entropy-balancing weights in the OLS models. In this subsection, we assess zeros, a high concentration of individuals with relatively few consultations, and a small number of observations with extremely high utilization the robustness of the main findings by examining the sensitivity of the 2SLS estimates to alternative measures of healthcare utilization and using an instrumental variable constructed without sampling weights. More specifically, we investigate whether the estimated causal effect is sensitive to the functional form adopted in the baseline specification, namely $\ln(\text{consultations}+1)$, as well as to the presence of a small number of extreme observations in the distribution of medical consultations.

As highlighted in the descriptive statistics, the number of consultations is characterized by a large proportion of zero values, a high concentration of individuals with relatively few consultations, and a small number of observations with extremely high utilization levels. For instance, the maximum reported number of consultations during the previous 12 months is 365. Such features may influence the estimated coefficients and raise concerns regarding the robustness of the results. To address this issue, we re-estimate the fully specified instrumental variable model by first replacing the weighted instrumental variable with an alternative measure constructed without sampling weights, defined as, $Z_i^{\text{unweighted}} = \frac{\sum_{j \in c, j \neq i} \text{Plano}_j}{N_c - 1}$. Second, we employ the Inverse Hyperbolic Sine (IHS) transformation of the number of consultations, $IHS(\text{consultations}_i) = \text{asinh}(\text{consultations}_i) = \ln\left(\text{consultations}_i + \sqrt{\text{consultations}_i^2 + 1}\right)$. This transformation accommodates zero values, reduces the influence of highly skewed observations, and preserves information from individuals with low utilization levels. Third, we construct a truncated version of the consultations variable by capping observations at the 99th percentile of the distribution to reduce the influence of extreme values. We then apply the logarithmic transformation to the truncated variable. The robustness checks are reported in Table 6. Our estimates show that the causal effect remains relatively stable across alternative specifications, yielding results that are relatively consistent with those obtained from the fully specified model presented in Column 5 of Table 3

Table 6: Robustness Checks for the Effect of Private Health Insurance on Annual Medical Consultations

	(1)	(2)	(3)
Plan	0.704*** (0.078)	0.913*** (0.102)	0.717*** (0.079)
State Fixed Effects	Yes	Yes	Yes
Observations	50514	50514	50514

Source: Authors' elaboration based on data from the 2019 National Health Survey (PNS). *Notes:* Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. All specifications include the full set of demographic, socioeconomic, chronic disease, and regional controls. For brevity, the estimated coefficients for these control variables are omitted from the table. The omitted reference groups are male, non White, individuals with no formal education, individuals without chronic conditions, rural residents, and residents of the state of Rondônia.

6.4 Heterogeneous Effects of Private Health Insurance on the Intensity of Health Care Utilization

Although the average effect of private health insurance estimated in this study is broadly consistent with previous evidence, the impact of insurance coverage may not be homogeneous across individuals. Private health insurance may increase the intensive utilization of health care services more strongly for some population groups than for others. In this section, we provide evidence that the effect of private health insurance varies according to gender, age, and health status. Estimating heterogeneous treatment effects can therefore contribute to a better understanding of the mechanisms through which private health insurance affects the intensity of health care utilization.

To estimate heterogeneous effects with respect to a given characteristic (H_i), we augment Equation 3 by including an interaction term between private health insurance coverage and the characteristic of interest. The coefficient associated with this interaction term captures the differential effect of private health insurance for individuals with characteristic (H_i). Because the interaction term is also endogenous by construction, we additionally include in the first-stage equation an interaction between the instrumental variable and (H_i). Consequently, the first-stage specification includes both the original instrument and the interaction between the instrument and the characteristic of interest.

Table 7 reports the second-stage 2SLS estimates. For each specification, we report the coefficients associated with the instrumented private health insurance variable (Plan), the characteristic (H_i), and the instrumented interaction term. The first column presents heterogeneous effects according to the presence of chronic health conditions. For this analysis, we construct a binary indicator equal to 1 if the individual has been diagnosed with at least one of the chronic diseases considered in this study, and 0 otherwise. Consequently, the vector of chronic disease indicators is excluded from this specification. The second column reports heterogeneous effects by gender, while the third column presents heterogeneous effects by age. For the age heterogeneity analysis the estimated effect for an individual of average age.

In Column 1, we observe that, among individuals without chronic diseases, the effect of private health insurance coverage is positive and statistically significant at the 0.1% level. The estimates indicate that private health insurance increases the number of medical consultations by approximately 135% $= 100(e^{0.855} - 1)$ for individuals without chronic conditions, whereas the corresponding causal effect for individuals with at least one chronic disease is approximately 67% $= 100(e^{0.855-0.342} - 1)$. The coefficient on the chronic disease indicator also reveals that, among individuals without private health insurance, having at least one chronic condition increases the number of medical consultations by approximately 59.5% $= 100(e^{0.467} - 1)$ relative to individuals without chronic diseases. These findings suggest that private health insurance generates substantially larger increases in health care utilization among individuals without chronic conditions than among those with at least one diagnosed chronic disease.

A plausible explanation for this result is that individuals with chronic conditions already make greater use of health care services, even in the absence of private health insurance coverage. This interpretation is consistent with the hybrid structure of the Brazilian health care system, in which public health services are available to patients with chronic diseases through the Unified Health System (SUS). Consequently, the marginal gain from obtaining private health insurance tends to be smaller for individuals with diagnosed chronic conditions. On the other hand, individuals without chronic diseases exhibit lower baseline levels of health care utilization. For this group, private health insurance may yield larger utilization gains by reducing access barriers commonly faced by those who rely exclusively on the public health care system.

Table 7: Heterogeneous Effects of Private Health Insurance on the Intensity of Health Care Utilization

ln(consultations+1)	(1)	(2)	(3)
Plan	0.855*** (0.089)	0.769*** (0.082)	0.699*** (0.081)
Plan*Chronic	-0.342*** (0.060)		
Chronic	0.467*** (0.022)		
Female	0.284*** (0.011)	0.313*** (0.020)	0.269*** (0.011)
Plan*Female		-0.133* (0.061)	
Plan*CenteredAge			-0.009** (0.003)
Centered Age			0.002** (0.001)
State Fixed Effects	Yes	Yes	Yes
Observations	50514	50514	50514

Source: Authors' elaboration based on data from the 2019 National Health Survey (PNS). *Notes:* Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. All specifications include the full set of demographic, socioeconomic, chronic disease, and regional controls. For brevity, the estimated coefficients for these control variables are omitted from the table.

Column 2 examines whether the effect of private health insurance differs between men and women. The estimated coefficients indicate that private health insurance has a larger effect on health care utilization among men, increasing the number of medical consultations by approximately $100(e^{0.855} - 1) = 115.8\%$, whereas the corresponding effect for women is $100(e^{0.855-0.113} - 1) = 88.9\%$. The coefficient on the interaction term is negative and statistically significant at the 5% level, indicating that the effect of private health insurance differs systematically by gender.

The results also show that, among individuals without private health insurance, women use medical consultations more intensively than men, and this difference is statistically significant. A plausible explanation for this finding is that women may exhibit a stronger preventive health care behavior than men, leading to higher baseline levels of health care utilization even in the absence of private insurance.

coverage. Consequently, because women already make greater use of health services, the marginal gain associated with obtaining private health insurance tends to be smaller than that observed for men.

This finding is consistent with international evidence. Using 22 years of longitudinal data, Green and Pope (1999) [42] document higher healthcare utilization among women even after controlling for important differences in health status, thereby testing the longstanding argument that women seek medical care more frequently simply because they experience higher morbidity. Their findings suggest that gender differences in healthcare utilization cannot be explained solely by differences in morbidity but are also associated with behavioral and attitudinal factors related to healthcare-seeking behavior.

Column 3 reports estimates of heterogeneous effects by age, centered at their sample means. The results indicate that, for an individual of average age, private health insurance increases the number of medical consultations by approximately 101.2%, calculated as $= 100(e^{0.699} - 1)$. The interaction coefficient is negative and statistically significant, indicating that this effect decreases by 0.009 log points for each additional year of age.

The coefficient on Centered Age also confirms previous evidence [43] that healthcare utilization increases with age among individuals without private health insurance. By contrast, the marginal effect of private health insurance declines with age, suggesting that insurance coverage yields larger utilization gains among younger individuals than among older ones. A plausible explanation is that older individuals already exhibit higher baseline levels of healthcare utilization due to their greater health needs and more frequent contact with the healthcare system. Consequently, private health insurance provides a smaller additional increase in utilization among older adults, whereas it may substantially reduce access barriers for younger individuals, leading to larger increases in medical consultations. This pattern is consistent with the heterogeneous effects previously identified for individuals with chronic conditions.

The corresponding first-stage estimates are reported in Appendix C, Table C3. Because the heterogeneous effects specification introduces a second endogenous variable and an additional instrument into the 2SLS framework, concerns regarding instrument relevance become particularly important. Table C3 reports the first-stage results for the heterogeneous-effects estimates. Columns 1 and 2 present the first-stage equations corresponding to the chronic condition heterogeneity specification, while the remaining columns report the first-stage results for the other heterogeneous effects models. Across nearly all specifications, the coefficients on both instruments are statistically significant, indicating a strong correlation with their respective endogenous variables. Moreover, the first-stage F-statistics consistently exceed the conventional threshold of 10, providing evidence against weak instrument concerns. Taken together, these results suggest that the instruments possess sufficient explanatory power to identify the heterogeneous effects of private health insurance coverage.

6.5 The Effect of Private Health Insurance on the Extensive Margin of Health Care Utilization

In the previous sections, we discussed how private health insurance affects healthcare utilization at the intensive margin, measured by the number of medical consultations during the 12 months preceding the National Health Survey (PNS). However, private health insurance may also affect the extensive margin of healthcare utilization, namely the individual's decision to seek medical care in the previous two weeks. Although both variables capture dimensions of healthcare utilization, they clearly reflect distinct behavioral responses to insurance coverage. While the first outcome captures the frequency with which individuals use healthcare services over time, the second reflects the initial decision to seek care. Therefore, analyzing the effect of private health insurance on seeking medical care during the previous two weeks is important because insurance coverage may reduce barriers to healthcare access. Consequently, private health insurance is expected not only to increase the frequency of healthcare utilization but also to raise the probability that individuals seek medical attention when facing a health need.

Using the same empirical strategy as in the previous sections, we estimate the effect of private health insurance on the probability of seeking medical care, with the results reported in Table 8. Controlling for demographic and socioeconomic characteristics, chronic conditions, and regional fixed effects, Column 1 presents OLS estimates from a linear probability model. Column 2 reports the marginal effects obtained from a logit specification estimated by maximum likelihood. Columns 3 and 4 present the 2SLS estimates using the instrumental variable constructed with and without survey weights, respectively, in order to assess the robustness of the instrumental variable results.

The estimates reported in Table 8 indicate that private health insurance significantly increases the probability of seeking medical care. The OLS estimates presented in Column 1 suggest that individuals covered by private health insurance are approximately 7 percentage points more likely to seek medical

care during the two weeks preceding the survey interview. Similarly, the marginal effects estimated from the logit specification in Column 2 indicate an increase of approximately 6.5 percentage points in the probability of seeking healthcare services.

The instrumental variable estimates reported in Columns 3 and 4 are larger in magnitude, suggesting that private health insurance increases the probability of seeking medical care by approximately 10.2 percentage points. The fact that the IV estimates exceed the OLS and logit estimates suggests the presence of downward bias in conventional estimators, potentially reflecting unobserved factors associated with lower healthcare-seeking behavior among individuals more likely to obtain private health insurance coverage. Moreover, the similarity between the IV estimates obtained with weighted and unweighted instruments reinforces the robustness of the results to alternative instrument constructions.

Overall, these findings indicate that private health insurance not only increases the intensity of healthcare utilization, as previously shown by the number of medical consultations, but also raises the likelihood that individuals seek medical attention when facing a health need. This result is consistent with the argument that private health insurance reduces barriers to healthcare access and facilitates entry into the healthcare system.

Table 8: The Effect of Private Health Insurance on the Extensive Margin of Healthcare Utilization

Health Care Seeking	(1)	(2)	(3)	(4)
Plan	0.070*** (0.008)	0.065*** (0.008)	0.102* (0.043)	0.102* (0.042)
Observations	50514	50514	50514	50514

Source: Authors' elaboration based on data from the 2019 National Health Survey (PNS). *Notes:* Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. All specifications include the full set of demographic, socioeconomic, chronic disease, and regional controls. For brevity, the estimated coefficients for these control variables are omitted from the table. The omitted reference groups are male, non-White individuals with no formal education, individuals without chronic conditions, rural residents, and residents of the state of Rondônia.

As in the analysis of the intensive margin of healthcare utilization, concerns about the validity of the instrumental variable also arise in the extensive-margin specification. However, the plausibility of the exogeneity assumption remains essentially the same as in the previous analysis, since the identification strategy and the instrument construction are unchanged. Moreover, because the first stage is identical to that estimated in the intensive-margin models, the instrument remains both relevant and sufficiently strong, as indicated by the first-stage results previously discussed. Taken together, these findings provide supportive evidence for the validity of the instrumental variable strategy adopted in this section.

7 Conclusions

This study aimed to provide new causal evidence on the effects of private health insurance on healthcare utilization in Brazil, examining both the extensive margin, captured by healthcare-seeking behavior over the previous two weeks, and the intensive margin, measured by the annual number of medical consultations. To address the endogenous nature of private health insurance coverage, arising from adverse selection and other sources of non-random treatment assignment, we employed an instrumental variables strategy based on the leave-one-out weighted share of private health insurance coverage within occupational-sector cells defined by sector of activity, formal employment status, and state of residence. The analysis draws on microdata from the 2019 National Health Survey (PNS), covering a sample of 50,514 employed individuals.

The OLS estimates consistently indicate a positive and statistically significant association between private health insurance and the number of medical consultations across all specifications. However, the sensitivity of the OLS coefficients to the sequential inclusion of observable controls, declining from 0.379 to 0.288 after the full set of covariates is added, suggests the presence of selection on observable characteristics. Even after balancing observable covariates through entropy balancing, the OLS estimates are likely to remain biased due to unobserved factors such as risk aversion, preventive behavior, and individual preferences for healthcare, which are correlated with both insurance status and healthcare utilization.

The IV estimates reveal that correcting for endogeneity substantially amplifies the estimated effect of private health insurance on healthcare utilization. The fully specified 2SLS model yields an estimated coefficient of 0.711, implying that private health insurance coverage more than doubles the expected annual number of medical consultations — an increase of approximately 103.6%. This result is consistent

with the hypothesis that uninsured individuals face significant barriers to healthcare access and unmet healthcare needs, which compress observed utilization and generate a negative bias in OLS estimates. The IV estimates remain robust across alternative specifications, including different transformations of the dependent variable and instrumental variable constructions that do not incorporate survey weights.

Regarding the first-stage estimates, the proposed instrument exhibits a strong and statistically significant association with private health insurance coverage across all specifications, with F-statistics well above the conventional weak-instrument thresholds. This evidence is consistent with the institutional structure of the Brazilian labor market, in which employer-sponsored health insurance is predominantly concentrated among formally employed workers in specific occupational sectors and geographic regions. The strong instrument relevance, combined with the plausibility of the exclusion restriction, which requires that occupational-level insurance diffusion affects healthcare utilization only through individual insurance coverage, supports the validity of the identification strategy.

One potential concern is that individuals may self-select into occupations or local environments associated with greater access to employer-provided health insurance and distinct healthcare utilization patterns. To mitigate this concern, our estimations control for an extensive set of demographic, socioeconomic, health, and regional characteristics that are likely correlated with both insurance coverage and healthcare. In addition, the leave-one-out construction of the instrumental variable reduces concerns related to mechanical correlation between the individual's insurance status and the instrument itself. Although the exclusion restriction cannot be tested directly, the stability of the results across alternative specifications and robustness checks provides supportive evidence for the validity of the identification strategy.

The results also reveal important heterogeneity in the causal effects of private health insurance. The impact on medical consultations is significantly larger among men, individuals without chronic diseases, and younger individuals. In particular, the marginal effect of health insurance on healthcare utilization declines with age, suggesting that private coverage generates stronger utilization gains among groups with lower baseline demand for medical services. Regarding the extensive margin, private health insurance moderately increases the probability of seeking healthcare services, while exerting a substantially stronger effect on the intensive margin by increasing the frequency of utilization conditional on access.

This study contributes to the health economics literature by providing new causal evidence on the effects of private health insurance on healthcare utilization in Brazil, a context where most existing studies rely on observational comparisons that do not account for the endogenous nature of insurance coverage. The instrumental variables strategy employed here exploits a distinct source of exogenous variation, occupational and regional heterogeneity in employer-sponsored insurance diffusion, extending the approach of Menezes-Filho and Politi (2020) and enabling estimation across the full distribution of employed workers. The findings carry important policy implications, suggesting that expanding access to private health insurance among underserved occupational groups could generate substantial gains in healthcare utilization. Nevertheless, some limitations should be acknowledged. The analysis is restricted to employed individuals, which limits the external validity of the findings to the broader population, including the unemployed and inactive. Additionally, the exclusion restriction, while theoretically grounded, cannot be formally tested, and potential violations arising from direct effects of occupational sorting on healthcare preferences cannot be fully ruled out. Future research could extend this framework to examine the effects of private health insurance on health outcomes and expenditures, as well as to investigate complementarities and substitution patterns between the supplementary and public healthcare sectors.

Appendix A Definition of Variables

Table A1: Definition of Variables

Category	Variable	Description
Dependent Variables	Medical Consultations	Number of medical consultations during the previous 12 months.
	Healthcare-Seeking Behavior	Dummy variable equal to 1 if the individual sought any healthcare service or professional for their own health during the previous two weeks, and 0 otherwise.
Treatment	Private Health Insurance	Dummy variable equal to 1 if the individual is covered by a private employer-sponsored or public-sector health insurance plan, and 0 otherwise.
Instrumental Variable	Occupational Health Insurance Coverage	Leave-one-out weighted share of individuals covered by private health insurance within the individual's occupational-sector cell, defined by sector of activity, formal employment status, and state of residence. The variable is weighted using individual sampling weights from the PNS.
Demographic Controls	Female	Dummy variable equal to 1 for females, and 0 otherwise.
	White	Dummy variable equal to 1 if the individual self-identifies as White, and 0 otherwise.
	Age Age Squared	Age in years. Squared term of age in years.
Socioeconomic Controls	Incomplete Primary Education	Dummy variable equal to 1 if the individual has incomplete primary education, and 0 otherwise.
	Complete Primary Education	Dummy variable equal to 1 if the individual has completed primary education, and 0 otherwise.
	Incomplete Secondary Education	Dummy variable equal to 1 if the individual has incomplete secondary education, and 0 otherwise.
	Complete Secondary Education	Dummy variable equal to 1 if the individual has completed secondary education, and 0 otherwise.
	Incomplete Higher Education	Dummy variable equal to 1 if the individual has incomplete higher education, and 0 otherwise.
	Complete Higher Education	Dummy variable equal to 1 if the individual has completed higher education, and 0 otherwise.
	Household Per Capita Income	Monthly household per capita income measured in Brazilian reais (BRL).
Chronic Conditions	Hypertension	Dummy variable indicating diagnosis of hypertension.
	Diabetes	Dummy variable indicating diagnosis of diabetes.
	High Cholesterol	Dummy variable indicating diagnosis of high cholesterol.
	Heart Disease	Dummy variable indicating diagnosis of heart disease.
	Stroke	Dummy variable indicating diagnosis of stroke.
	Asthma	Dummy variable indicating diagnosis of asthma.
	Arthritis	Dummy variable indicating diagnosis of arthritis.
	Back Pain	Dummy variable indicating chronic back pain condition.
	Depression	Dummy variable indicating diagnosis of depression.
	Pulmonary Disease	Dummy variable indicating diagnosis of chronic pulmonary disease.
Regional Controls	Cancer	Dummy variable indicating diagnosis of cancer.
	Kidney Disease	Dummy variable indicating diagnosis of kidney disease.
	Urban	Dummy variable equal to 1 for urban residents, and 0 otherwise.
	State Fixed Effects	State-level dummy variables.

Notes: Individuals with missing information for the number of medical consultations, but who reported that their last medical appointment occurred more than one year prior to the survey or that they had never had a medical consultation, were assigned a value of zero consultations. The occupational-sector cells used to construct the instrumental variable are defined by sector of activity, formal employment status, and state of residence. Cells with fewer than 20 observations were excluded from the analysis.

All chronic condition variables are binary indicators based on self-reported medical diagnosis.

Appendix B Balance

Table B2: Covariate Balance Before and After Balancing

Covariates	Before Balancing						After Balancing					
	Treat			Control			Treat			Control		
	Mean	Variance	Skewness	Mean	Variance	Skewness	Mean	Variance	Skewness	Mean	Variance	Skewness
Female	0.487	0.250	0.051	0.420	0.244	0.323	0.487	0.250	0.051	0.487	0.250	0.051
White	0.529	0.249	-0.117	0.315	0.216	0.798	0.529	0.249	-0.117	0.529	0.249	-0.117
Age	43.070	167.100	0.371	41.910	180.400	0.282	43.070	167.100	0.371	43.070	167.100	0.268
Age Squared	2022	1423621	1.031	1937	1430768	0.959	2022	1423621	1.031	2022	1380363	0.895
Incomplete Primary Educ.	0.070	0.065	3.361	0.329	0.221	0.727	0.070	0.065	3.361	0.070	0.065	3.361
Complete Primary Educ.	0.038	0.036	4.861	0.088	0.081	2.900	0.038	0.036	4.861	0.038	0.036	4.861
Incomplete Secondary Educ.	0.032	0.031	5.346	0.079	0.073	3.123	0.032	0.031	5.346	0.032	0.031	5.346
Complete Secondary Educ.	0.279	0.201	0.985	0.306	0.212	0.842	0.279	0.201	0.985	0.279	0.201	0.985
Incomplete Higher Educ.	0.080	0.073	3.104	0.042	0.040	4.554	0.080	0.073	3.104	0.080	0.073	3.104
Complete Higher Educ.	0.494	0.250	0.025	0.103	0.092	2.622	0.494	0.250	0.025	0.494	0.250	0.025
Ln(PC Household Income)	7.788	0.825	0.340	6.686	0.746	-0.165	7.788	0.825	0.340	7.788	0.850	0.431
Hypertension	0.199	0.159	1.509	0.181	0.148	1.657	0.199	0.159	1.509	0.199	0.159	1.509
Diabetes	0.052	0.049	4.037	0.050	0.048	4.108	0.052	0.049	4.037	0.052	0.049	4.037
High Cholesterol	0.160	0.134	1.858	0.106	0.095	2.554	0.160	0.134	1.858	0.160	0.134	1.858
Heart Disease	0.039	0.037	4.784	0.029	0.028	5.625	0.039	0.037	4.784	0.039	0.037	4.784
Stroke	0.008	0.008	11.110	0.010	0.010	9.875	0.008	0.008	11.110	0.008	0.008	11.110
Asthma	0.063	0.059	3.599	0.038	0.036	4.857	0.063	0.059	3.599	0.063	0.059	3.599
Arthritis	0.055	0.052	3.912	0.046	0.044	4.321	0.055	0.052	3.912	0.055	0.052	3.912
Chronic Back Pain	0.034	0.033	5.136	0.018	0.017	7.324	0.034	0.033	5.136	0.034	0.033	5.136
Depression	0.102	0.092	2.624	0.066	0.062	3.494	0.102	0.092	2.624	0.102	0.092	2.624
Pulmonary Disease	0.012	0.011	9.126	0.009	0.009	10.470	0.012	0.011	9.126	0.012	0.011	9.126
Cancer	0.027	0.026	5.880	0.012	0.012	8.840	0.027	0.026	5.880	0.027	0.026	5.880
Kidney Disease	0.009	0.009	10.500	0.011	0.011	9.464	0.009	0.009	10.500	0.009	0.009	10.500
Urban	0.945	0.052	-3.902	0.733	0.196	-1.054	0.945	0.052	-3.902	0.945	0.052	-3.901

Source: Authors' elaboration based on data from the 2019 National Health Survey (PNS). *Note:* Coefficients for state dummy variables (UF fixed effects) are not reported due to space constraints.

Appendix C Heterogeneous Effects

Table C3: First Stage Results: Heterogeneous Effects Specifications

	(1) plano	(2) plano_cronico	(3) plano	(4) plano_mulher	(5) plano	(6) plano_idade
Chronic	0.019 ⁺ (0.010)	0.041*** (0.008)				
Female	-0.002 (0.007)	-0.014** (0.005)	0.018* (0.009)	0.033*** (0.007)	-0.003 (0.007)	-0.158 (0.098)
Age	0.004** (0.001)	-0.002* (0.001)	0.004** (0.001)	0.002* (0.001)		
Age Squared	-0.000* (0.000)	0.000* (0.000)	-0.000* (0.000)	-0.000* (0.000)		
Z	0.446*** (0.027)	-0.196*** (0.010)	0.504*** (0.028)	-0.202*** (0.011)	0.476*** (0.023)	0.450 (0.335)
Z*Chronic	0.068* (0.033)	0.972*** (0.026)				
Z*female			-0.068* (0.034)	0.897*** (0.027)		
Centered Age					0.000 (0.000)	0.041*** (0.007)
Centered Age Squared					-0.000* (0.000)	0.004*** (0.000)
Z*CenteredAge					0.003 ⁺ (0.001)	0.873*** (0.026)
State Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
F Statistic	269.95	182.87	224.56	139.20	225.82	58.46
Observations	50514	50514	50514	50514	50514	50514

Source: Authors' elaboration based on data from the 2019 National Health Survey (PNS). *Notes:* Standard errors in parentheses.
⁺ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Coefficients for the control variables are omitted due to space constraints.

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